

```
import pandas as pd
```

```
# Load customer dataset
```

```
df = pd.read_csv("Customer_Transactions.csv")
```

```
# Display first 5 records
```

```
print(df.head())
```

	Customer_ID	Age	Gender	Annual_Income	Purchase_Frequency
0	C001	22	Female	25000	5
1	C002	35	Male	50000	12
2	C003	28	Female	32000	7
3	C004	45	Male	70000	20
4	C005	31	Female	40000	10

	Preferred_Category
0	Fashion
1	Electronics
2	Beauty
3	Electronics
4	Grocery

```
# Check missing values in each column
```

```
print(df.isnull().sum())
```

Customer_ID	0
Age	0
Gender	0
Annual_Income	0
Purchase_Frequency	0
Total_Spending	0
Preferred_Category	0

dtype: int64

```
# Fill missing values only in numeric columns
```

```
df['Age'] = df['Age'].fillna(df['Age'].mean())
```

```
df['Annual_Income'] =
```

```
df['Annual_Income'].fillna(df['Annual_Income'].mean())
```

```
df['Purchase_Frequency'] =
```

```
df['Purchase_Frequency'].fillna(df['Purchase_Frequency'].mean())
```

```
df['Total_Spending'] =
```

```
df['Total_Spending'].fillna(df['Total_Spending'].mean())
```

```
print(df.isnull().sum())
```

```
Customer_ID      0
Age              0
Gender           0
Annual_Income    0
Purchase_Frequency 0
Total_Spending   0
Preferred_Category 0
dtype: int64
```

```
# Convert Gender column into numbers
# Male = 1, Female = 0
```

```
df['Gender'] = df['Gender'].map({
    'Male':1,
    'Female':0
})
```

```
print(df.head())
```

	Customer_ID	Age	Gender	Annual_Income	Purchase_Frequency
0	C001	22	0	25000	5
1	C002	35	1	50000	12
2	C003	28	0	32000	7
3	C004	45	1	70000	20
4	C005	31	0	40000	10

	Preferred_Category
0	Fashion
1	Electronics
2	Beauty
3	Electronics
4	Grocery

```
from sklearn.preprocessing import MinMaxScaler
```

```
# Select columns for normalization
```

```
columns = ['Annual_Income', 'Purchase_Frequency', 'Total_Spending']
```

```
# Create scaler object
```

```
scaler = MinMaxScaler()
```

```
# Apply normalization
```

```
df[columns] = scaler.fit_transform(df[columns])
```

```
print(df.head())
```

	Customer_ID	Age	Gender	Annual_Income	Purchase_Frequency
Total_Spending					
0	C001	22	0	0.000000	0.047619
0.022222					
1	C002	35	1	0.416667	0.380952
0.388889					
2	C003	28	0	0.116667	0.142857
0.088889					
3	C004	45	1	0.750000	0.761905
0.777778					
4	C005	31	0	0.250000	0.285714
0.277778					

	Preferred_Category
0	Fashion
1	Electronics
2	Beauty
3	Electronics
4	Grocery

```

from sklearn.cluster import KMeans

# Features used for clustering
X = df[['Annual_Income',
        'Purchase_Frequency',
        'Total_Spending']]

# Create K-Means model with 3 clusters
kmeans = KMeans(n_clusters=3, random_state=42)

# Fit model and assign clusters
df['Cluster'] = kmeans.fit_predict(X)

# Display cluster results
print(df[['Customer_ID', 'Cluster']])

```

	Customer_ID	Cluster
0	C001	0
1	C002	1
2	C003	0
3	C004	2
4	C005	1
5	C006	0
6	C007	2
7	C008	2
8	C009	0
9	C010	1
10	C011	0
11	C012	2
12	C013	1

```
13      C014      0
14      C015      1
```

```
# Display customers and their clusters
```

```
for i in range(3):
    print("\nCluster", i)
    print(df[df['Cluster']==i])
```

Cluster 0

	Customer_ID	Age	Gender	Annual_Income	Purchase_Frequency	\
0	C001	22	0	0.000000	0.047619	
2	C003	28	0	0.116667	0.142857	
5	C006	24	1	0.050000	0.000000	
8	C009	27	0	0.083333	0.095238	
10	C011	29	0	0.166667	0.190476	
13	C014	26	1	0.033333	0.047619	

	Total_Spending	Preferred_Category	Cluster
0	0.022222	Fashion	0
2	0.088889	Beauty	0
5	0.000000	Fashion	0
8	0.055556	Fashion	0
10	0.133333	Beauty	0
13	0.033333	Fashion	0

Cluster 1

	Customer_ID	Age	Gender	Annual_Income	Purchase_Frequency	\
1	C002	35	1	0.416667	0.380952	
4	C005	31	0	0.250000	0.285714	
9	C010	41	1	0.583333	0.523810	
12	C013	33	0	0.333333	0.333333	
14	C015	37	0	0.500000	0.476190	

	Total_Spending	Preferred_Category	Cluster
1	0.388889	Electronics	1
4	0.277778	Grocery	1
9	0.555556	Grocery	1
12	0.333333	Grocery	1
14	0.500000	Beauty	1

Cluster 2

	Customer_ID	Age	Gender	Annual_Income	Purchase_Frequency	\
3	C004	45	1	0.750000	0.761905	
6	C007	39	0	0.666667	0.666667	
7	C008	52	1	1.000000	1.000000	
11	C012	48	1	0.883333	0.857143	

	Total_Spending	Preferred_Category	Cluster
--	----------------	--------------------	---------

3	0.777778	Electronics	2
6	0.666667	Beauty	2
7	1.000000	Electronics	2
11	0.888889	Electronics	2

Calculate average values for each cluster

```
cluster_summary = df.groupby('Cluster')[
    ['Annual_Income',
     'Purchase_Frequency',
     'Total_Spending']
].mean()
```

```
print(cluster_summary)
```

Cluster	Annual_Income	Purchase_Frequency	Total_Spending
0	0.075000	0.087302	0.055556
1	0.416667	0.400000	0.411111
2	0.825000	0.821429	0.833333

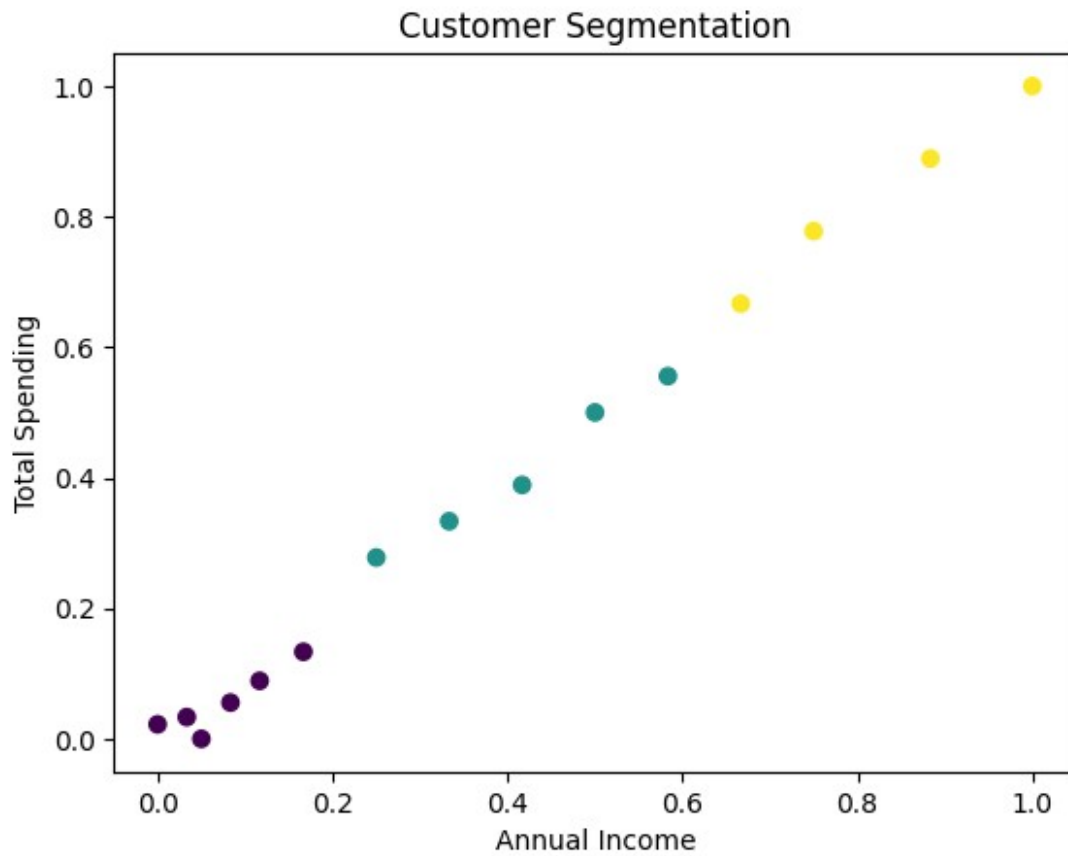
```
import matplotlib.pyplot as plt
```

Scatter plot of customer segments

```
plt.scatter(
    df['Annual_Income'],
    df['Total_Spending'],
    c=df['Cluster']
)
```

```
plt.xlabel("Annual Income")
plt.ylabel("Total Spending")
plt.title("Customer Segmentation")
```

```
plt.show()
```



Business Insights

- 1.High-value customers spend more and help increase sales.
- 2.Low-value customers may buy more if given discounts.
- 3.Medium-value customers can become loyal customers.
- 4.Electronics customers spend more than others.
- 5.Customer groups help the company give better offers.